Scale 5 Fractional Factorial Design

Eftychis Kontadakis

Introduction

We are going to project a 4 factor factorial design (A,B,C,D) to 3 factor fractional design of scale 4

The Data

First thing's first, we insert the data. All of them.

```r
Y<- c(18,19,44,62,26,32,55,70,16,20,40,60,25,31,54,73)
#(1),a,b,ab,c,ac,bc,abc,d,ad,bd,abd,cd,acd,bcd,abcd
```

We are going to create the full design matrix:

```r
D   <- rep(c(-1,1), each = 8)
C   <- rep(rep(c(-1,1), each = 4), 2)
B   <- rep(rep(c(-1,1), each = 2), 4)
A   <- rep(rep(c(-1,1), each = 1), 8)
AB  <- A*B
AC  <- A*C
AD  <- A*D
BC  <- C*B
BD  <- B*D
CD  <- C*D
ABC <- A*C*B
ABD <- A*B*D
ACD <- A*C*D
BCD <- B*C*D
ABCD<- A*B*C*D
#We create the design matrix
Design <- data.frame(
```

```r
  Y    = Y,
  Intercept = 1,
  A    = A,
  B    = B,
  C    = C,
  D    = D,
  AB   = AB,
  AC   = AC,
  AD   = AD,
  BC   = BC,
  BD   = BD,
  CD   = CD,
  ABC  = ABC,
  ABD  = ABD,
  ACD  = ACD,
  BCD  = BCD,
  ABCD= ABCD
)
Design
```

```
    Y Intercept  A  B  C  D AB AC AD BC BD CD ABC ABD ACD BCD ABCD
1  18         1 -1 -1 -1 -1  1  1  1  1  1  1  -1  -1  -1  -1    1
2  19         1  1 -1 -1 -1 -1 -1 -1  1  1  1   1   1   1  -1   -1
3  44         1 -1  1 -1 -1 -1  1  1 -1 -1  1   1   1  -1   1   -1
4  62         1  1  1 -1 -1  1 -1 -1 -1 -1  1  -1  -1   1   1    1
5  26         1 -1 -1  1 -1  1 -1  1 -1  1 -1   1  -1   1   1   -1
6  32         1  1 -1  1 -1 -1  1 -1 -1  1 -1  -1   1  -1   1    1
7  55         1 -1  1  1 -1 -1 -1  1  1 -1 -1  -1   1   1  -1    1
8  70         1  1  1  1 -1  1  1 -1  1 -1 -1   1  -1  -1  -1   -1
9  16         1 -1 -1 -1  1  1  1 -1  1 -1 -1  -1   1   1   1   -1
10 20         1  1 -1 -1  1 -1 -1  1  1 -1 -1   1  -1  -1   1    1
11 40         1 -1  1 -1  1 -1  1 -1 -1  1 -1   1  -1   1  -1    1
12 60         1  1  1 -1  1  1 -1  1 -1  1 -1  -1   1  -1  -1   -1
13 25         1 -1 -1  1  1  1 -1 -1 -1 -1  1   1   1  -1  -1    1
14 31         1  1 -1  1  1 -1  1  1 -1 -1  1  -1  -1   1  -1   -1
15 54         1 -1  1  1  1 -1 -1 -1  1  1  1  -1  -1  -1   1   -1
16 73         1  1  1  1  1  1  1  1  1  1  1   1   1   1   1    1
```

These are the lines of this design matrix that satisfy the generator I=ABCDE, or E=ABCD:

Now I run the ANOVA:

```r
# We turn our data into a data frame
df2 <- data.frame(
  Y = Design$Y,
  A = Design$A,
  B = Design$B,
  C = Design$C,
  D = Design$D
)

model2 <- lm(Y ~ A*B*C*D, data = df2)
summary(model2)
```

```

Call:
lm(formula = Y ~ A * B * C * D, data = df2)

Residuals:
ALL 16 residuals are 0: no residual degrees of freedom!

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.3125        NaN     NaN      NaN
A             5.5625        NaN     NaN      NaN
B            16.9375        NaN     NaN      NaN
C             5.4375        NaN     NaN      NaN
D            -0.4375        NaN     NaN      NaN
A:B           3.4375        NaN     NaN      NaN
A:C           0.1875        NaN     NaN      NaN
B:C           0.3125        NaN     NaN      NaN
A:D           0.5625        NaN     NaN      NaN
B:D          -0.0625        NaN     NaN      NaN
C:D           0.4375        NaN     NaN      NaN
A:B:C        -0.6875        NaN     NaN      NaN
A:B:D         0.1875        NaN     NaN      NaN
A:C:D        -0.0625        NaN     NaN      NaN
B:C:D         0.5625        NaN     NaN      NaN
A:B:C:D       0.3125        NaN     NaN      NaN

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared:      1, Adjusted R-squared:    NaN
F-statistic:   NaN on 15 and 0 DF,  p-value: NA
```

It is clear that we can only calculate all the main effects and all of the 2-factor interactions.

Now we want to display the estimates again but in a column from the biggest absolute value to the lowest:

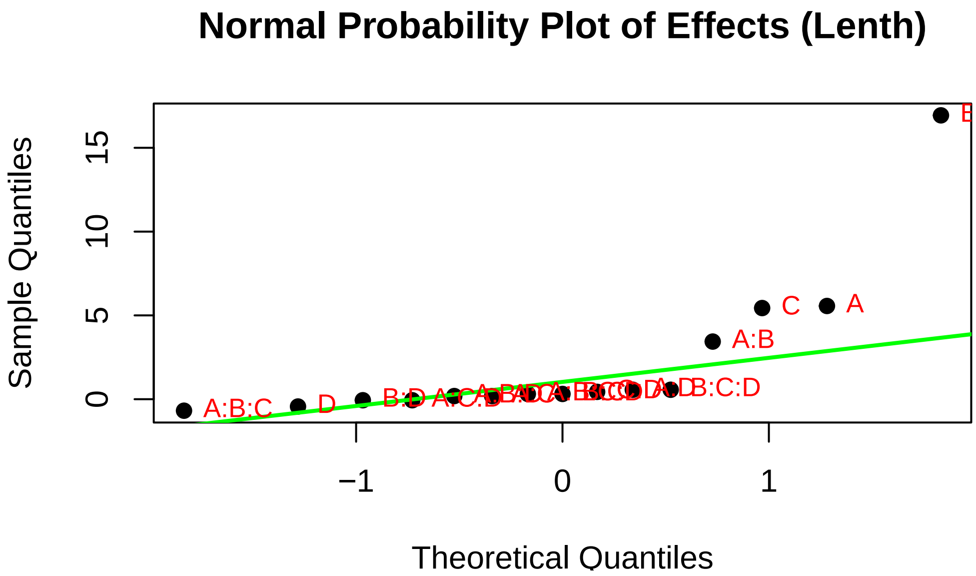
```
# Get the estimated coefficients (exclude the intercept)
effects <- coef(model2)[-1]

# Sort by absolute value, descending
effects_sorted <- effects[order(-abs(effects))]

# Display
effects_sorted
```

```
      B       A       C     A:B   A:B:C     A:D   B:C:D     C:D       D     B:C
16.9375  5.5625  5.4375  3.4375 -0.6875  0.5625  0.5625  0.4375 -0.4375  0.3125
A:B:C:D     A:C   A:B:D     B:D   A:C:D
 0.3125  0.1875  0.1875 -0.0625 -0.0625
```

We can clearly see that B is important, as well as A and C. We can also see that AB is important.

To finish we are going to calculate a Pseudo-Standard-Error(PSE) to calculate estimated t-values to be certain about what is important and what is not:

```
         Factor Estimate     PSE    t_value
B             B  16.9375 0.65625 25.8095238
A             A   5.5625 0.65625  8.4761905
C             C   5.4375 0.65625  8.2857143
A:B         A:B   3.4375 0.65625  5.2380952
A:B:C     A:B:C  -0.6875 0.65625 -1.0476190
A:D         A:D   0.5625 0.65625  0.8571429
B:C:D     B:C:D   0.5625 0.65625  0.8571429
C:D         C:D   0.4375 0.65625  0.6666667
D             D  -0.4375 0.65625 -0.6666667
B:C         B:C   0.3125 0.65625  0.4761905
A:B:C:D A:B:C:D   0.3125 0.65625  0.4761905
A:C         A:C   0.1875 0.65625  0.2857143
A:B:D     A:B:D   0.1875 0.65625  0.2857143
B:D         B:D  -0.0625 0.65625 -0.0952381
A:C:D     A:C:D  -0.0625 0.65625 -0.0952381
```



It is also visible from this plot that these are the important factors.